

Model-Independent Description of Multicomponent Adsorption from Affinity Spectrum Statistics



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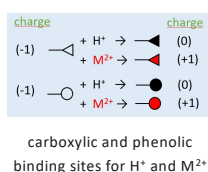
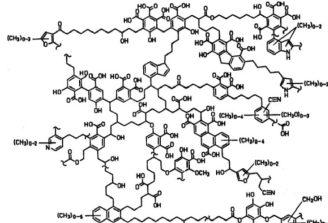
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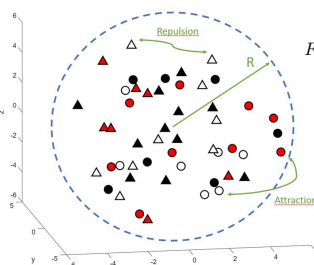
1. Motivation

Multi-cation adsorption to natural organic matter is described by an **affinity spectrum** — a density of binding free energies over all sites — showing *intrinsic* and *electrostatically induced* heterogeneity [1,2]



2. Monte Carlo model of randomly distributed sites

Semi-grand Canonical Ensemble: proton/metal occupation numbers fluctuate at fixed H and M chemical potentials; sites interact via a screened Coulomb term:



$$F(\{s_i, m_i\}) = - \sum_{i=1}^n \mu_{H,i} s_i - \sum_{i=1}^n \mu_{M,i} m_i - \sum_{j>i}^n \phi_{ij}^{\text{int}}$$

$$\mu_{H,i} = \ln(10) (pH - pK_{a,i})$$

$$\mu_{M,i} = \ln(10) (pM - pK_{M,i})$$

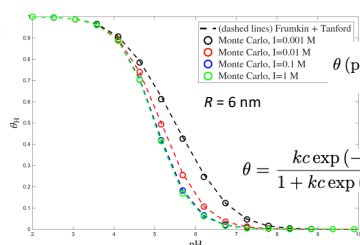
$$\phi_{ij}^{\text{int}} = \frac{q_i q_j \ell_B}{r_{ij}} \exp(-\kappa r_{ij}) \quad ; \quad q_i = 0, \pm 1, \pm 2, \dots$$

3. Mono-component Affinity Spectrum

Frumkin isotherm describes both intrinsic and electrostatically induced heterogeneity of MC data. No fitting is required with the Tanford interaction energy [3]:

$$p_{\text{Tanford}} = 3n \frac{\ell_B}{R} \left(\frac{1}{y^2} - \left(\frac{3}{2y^5} \right) (y^2 - 1 + (1+y)^2 \exp(-y)) \right)$$

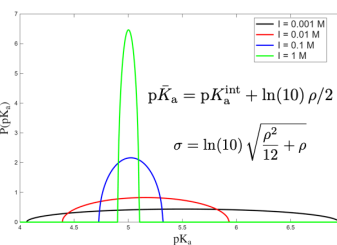
$$y = R/\ell_D$$



$$\theta(pH) = \int_{-\infty}^{\infty} P(pK_a) \frac{1}{1 + 10^{pH - pK_a}} dpK_a$$

$P(pK_a)$ spectra are recovered from $\theta(pH)$ by analytical inversion [4,5]

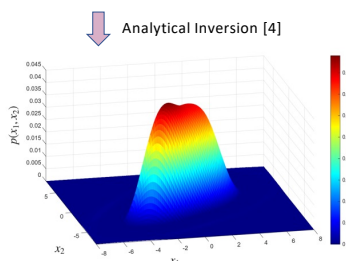
$$P(pK_a) = \frac{\ln(10)}{\pi} |\text{Im}[\theta(-10^{pK_a})]|$$



4. Multi-component Affinity Spectrum (Frumkin)

$$\theta_1(c_1, c_2) = \frac{k_1 c_1 \exp(-\rho_{11} \theta_1 - \rho_{12} \theta_2)}{1 + k_1 c_1 \exp(-\rho_{11} \theta_1 - \rho_{12} \theta_2) + k_2 c_2 \exp(-\rho_{21} \theta_1 - \rho_{22} \theta_2)}$$

$$\theta_2(c_1, c_2) = \frac{k_2 c_2 \exp(-\rho_{21} \theta_1 - \rho_{22} \theta_2)}{1 + k_1 c_1 \exp(-\rho_{11} \theta_1 - \rho_{12} \theta_2) + k_2 c_2 \exp(-\rho_{21} \theta_1 - \rho_{22} \theta_2)}$$



For any given first and second moments of the spectrum (mean $\bar{x}_i$, marginal widths σ_{ii} , and correlation coefficient R_C) there is always an associated Frumkin isotherm:

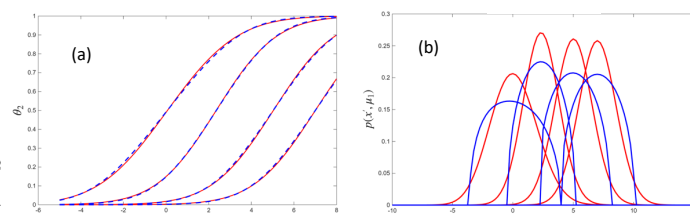
$$\rho_{ii} = -1 + \sqrt{1 + \frac{\sigma_{ii}^2}{3}} \quad \ln k_i = -\bar{x}_i + \rho_{ii}/2$$

$$\rho_{12} = \frac{(\rho_{11} + \rho_{22} - \bar{\rho})}{2} \quad \text{with} \quad \bar{\rho} = -1 + \sqrt{1 + \frac{\bar{\sigma}^2}{3}}$$

$$\text{and} \quad \bar{\sigma}^2 = \left(\frac{\sigma_{11}^2}{\sigma_{22}^2} + \frac{\sigma_{22}^2}{\sigma_{11}^2} - 2R_C \right) \sigma_{11} \sigma_{22}$$

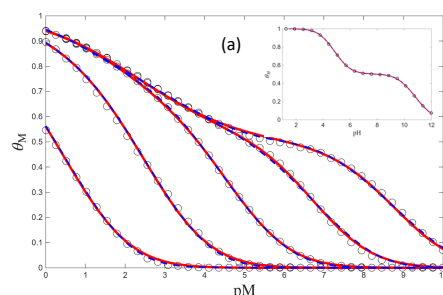
5. Primacy of the lowest cumulants

Only the lowest cumulants matter: affinity spectra with the same average and variance give almost identical isotherms



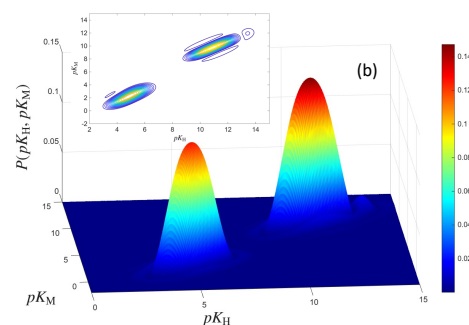
Blue = Frumkin isotherm; Red = isotherm from Gaussian spectrum, both with $\sigma_{11} = \sigma_{22} = 2$, $R_C = 0.7$, $x_1 = x_2 = 0$ and fixed $\mu_1 = -4, 2, 5, 7$; (a) Isotherms (b) Conditional Affinity Spectrum [5]. Note that the Gaussian and Frumkin isotherms almost overlap

6. Monte Carlo validation (bimodal distribution)



Fitting results (Frumkin and Gaussian models):

	Mode 1 (Frum.)	Mode 1 (Gauss.)	Mode 2 (Frum.)	Mode 2 (Gauss.)
pK_a^{av}	4.91	4.90	10.88	10.88
pK_M^{av}	2.26	2.27	9.73	9.68
σ_H	0.68	0.70	0.78	0.83
σ_M	1.75	1.98	1.70	1.80
R_C	0.85	0.90	0.98	0.96
Q_i	0.49	0.50	0.50	0.49



(a) Markers: MC simulations of metal titration curves at fixed pH = 2, 4, 6, 8, 10. $I = 0.01$ M, $pK_a^{\text{int}} = 4.5$, $pK_M^{\text{int}} = 4$, (abundance $Q_1 = 0.5$); $pK_a^{\text{int}} = 9.6$, $pK_M^{\text{int}} = 8$, $R = 4$ nm ($Q_2 = 0.5$). Inset: proton titration. Blue/red=best-fit bimodal Frumkin and Gaussian models. (b) Bimodal Frumkin competitive spectrum: electrostatics induce positive H-M binding correlation

Take-home message

- The lowest cumulants of the affinity spectrum (mean and covariance matrix) capture most of the information provided by the binding isotherms, as illustrated with multicomponent Gaussian and Frumkin spectra. This suggests a model-independent way to standardize experimental binding data.
- The Frumkin isotherm reproduces multicomponent MC simulations for both intrinsic and electrostatic heterogeneities. Its underlying affinity spectrum has a well-defined mean and covariance, and does not require numerical calculation of multiple integrals —unlike the Gaussian model.

[1] Gledhill, M., Gosnell, K., Humphreys, M. P., Delaigue, L., Helle, N., Zhu, K., Lodeiro, P., Rey-Castro, C., Achterberg E. P. (2026) *Nature Communications* 17, 3533; [2] Lodeiro, P., Rey-Castro, C., David, C., Humphreys, M. P., Gledhill, M. (2023). *Environ. Sci. Technol.*, 57 (50), 21136-21144; [3] Tanford, C. (1961). *Physical Chemistry of Macromolecules*. New York: John Wiley & Sons, Chapter 7; [4] Garcés J. L., Mas, F., Puy J. (2004) *J. Chem. Phys.* 120 (19) 9266-76; [5] Garcés J. L., Mas, F., Puy J. (2006) *J. Chem. Phys.* 124, 044710



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